Unit-4

Three Dimensional Geometry

Equation of a sphere :-

(i) If centre is origin and r is radius, then equation of sphere is given as

$$\boxed{x^2+y^2+z^2=r^2}$$

(ii) If centre is a, b, c and with radius 'r'

$$(x-a)^2+(y-b)^2+(z-c)^2=r^2$$

(iii) The general form of a sphere is given by

$x^2+y^2+z^2+2ux+2vy+2wz+d=0$ where u, v, w and d are four constant of the general equation of the sphere. It will be find out.

→ The centre of a sphere is $=(-u,-v,-w)$

radius $=\sqrt{u^2+v^2+w^2-d}$

(Q) Find the centre and radius of sphere for the equation $6(x^2+y^2+z^2)-8x+10y-18z-13=0$

Sol^n:- $\Rightarrow x^2+y^2+z^2-\frac{8}{6}x+\frac{10}{6}y-\frac{18}{6}z-\frac{13}{6}=0$

$\Rightarrow x^2+y^2+z^2-\frac{4}{3}x+\frac{10}{6}y-\frac{18}{6}z-\frac{13}{6}=0$

$\Rightarrow x^2+y^2+z^2+2\cdot\left(\frac{-2}{3}\right)x+2\left(\frac{5}{6}\right)y+2\left(\frac{-9}{6}\right)+\left(\frac{-13}{6}\right)=0$

Here, $u=\frac{-2}{3}$, $v=\frac{5}{6}$, $w=\left(\frac{-9}{6}\right)=\frac{-3}{2}$, $d=\frac{-13}{6}$

$\Rightarrow$ Centre of sphere $=(-u,-v,-w)$

$=\left(\frac{2}{3},\frac{-5}{6},\frac{3}{2}\right)$

→ radius $=\sqrt{\left(\frac{-2}{3}\right)^2+\left(\frac{5}{6}\right)^2+\left(\frac{-3}{2}\right)+\frac{13}{6}}$

$$= \sqrt{\frac{4}{9} + \frac{25}{36} + \frac{9}{4} + \frac{13}{6}}$$

$$= \sqrt{\frac{4\times4}{9\times4} + \frac{25}{36} + \frac{9\times9}{4\times9} + \frac{13\times6}{6\times6}}$$

$$= \sqrt{\frac{16+25+81+78}{36}}$$

$$= \sqrt{\frac{200}{36}}$$

$$= \frac{14.14}{6} \quad = \sqrt{5.55}$$

$$\boxed{= 2.35} \quad \boxed{= \frac{5}{3}\sqrt{2}}$$

(Q) find a eq^n of sphere through the four point $(0,0,0)$, $(-a,b,c)$, $(a,-b,c)$, $(a,b,-c)$ and find its radius also.

→ The general eq^n of sphere is

$$x^2+y^2+z^2+2ux+2vy+2wz+d=0 \quad -(1)$$

→ at point $(0,0,0)$ eq^n (1) become

$$d=0 \quad -(2)$$

→ similarly at $(-a,b,c)$.

$$a^2+b^2+c^2-2au+2bv+2cw=0 \quad -(3)$$

→ similarly at $(a,-b,c)$

$$a^2+b^2+c^2+2au-2bv+2cw=0 \quad -(4)$$

→ similarly at $(a,b,-c)$

$$a^2+b^2+c^2+2au+2bv-2cw=0 \quad -(5)$$

→ adding eq^n ③ and ④

$$2(a^2+b^2+c^2)+4wc=0$$

$$\boxed{w=\frac{-(a^2+b^2+c^2)}{2c}}$$

→ adding ④ and ⑤

$$2(a^2+b^2+c^2)+4au=0$$

$$u=-\frac{(a^2+b^2+c^2)}{2a}$$

→ adding ⑤ and ③

$$2(a^2+b^2+c^2)+4bv=0$$

$$v=-\frac{(a^2+b^2+c^2)}{2b}$$

put value of w, u, and v in eq^n ①

Ans:- $$x^2+y^2+z^2+2\left(\frac{-(a^2+b^2+c^2)}{2a}x\right)+2\left(\frac{-(a^2+b^2+c^2)}{2b}\right)y+2\left(\frac{-(a^2+b^2+c^2)}{2c}\right)z+0=0$$

$$r=\sqrt{\left(\frac{(a^2+b^2+c^2)^2}{4a^2}\right)+\frac{(a^2+b^2+c^2)^2}{4b^2}+\frac{(a^2+b^2+c^2)^2}{4c^2}-0}$$

Ans:- $$\boxed{r=\left(\frac{a^2+b^2+c^2)^{3/2}}{2abc}\right.}$$

Date
Page

Q. A plane passing through a fixed point (a,b,c) and its cut the axis (A,B,C) with respectively. show that locus of the centre of the sphere. OABC is $\frac{a}{x}+\frac{b}{y}+\frac{c}{z}=2$.

Sol^n:- The general eq^n of sphere is

$$x^2+y^2+z^2+2ux+2vy+2wz+d=0 \quad -(1)$$

Let the plane meet at the point $(0,0,0)$, $(\alpha,0,0)$, $(0,\beta,0)$, $(0,0,\gamma)$

→ at point $(0,0,0)$

$$d=0 \quad -(2)$$

→ at point $(\alpha,0,0)$

$$\alpha^2+2\alpha u=0 \quad -(3) \qquad u=\frac{-\alpha^2}{2\alpha}=\frac{-\alpha}{2}$$

→ at point $(0,\beta,0)$

$$\beta^2+2\beta v=0 \quad -(4) \qquad v=\frac{-\beta^2}{2\beta}=\frac{-\beta}{2}$$

→ at point $(0,0,\gamma)$

$$\gamma^2+2\gamma w=0 \quad -(5) \qquad w=\frac{-\gamma^2}{2\gamma}=\frac{-\gamma}{2}$$

put value of u, v and w in eq^n-(1)

$$x^2+y^2+z^2-\alpha x-\beta y-\gamma z=0 \quad -(6)$$

→ Now the eq^n of plane is $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=1$

at fixed point (a,b,c)

$$\frac{a}{\alpha}+\frac{b}{\beta}+\frac{c}{\gamma}=1 \quad - (7)$$

Let (x,y,z) be the centre of sphere of eq^n (6)

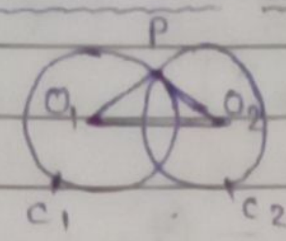
$$\frac{\alpha}{2}=x, \quad \frac{\beta}{2}=y, \quad \frac{\gamma}{2}=z$$

$$\alpha=2x, \quad \beta=2y, \quad \gamma=2z$$

putting this eq^n in (7)

$$\boxed{\frac{a}{x}+\frac{b}{y}+\frac{c}{z}=2}$$ Hence proved.

Intersection of sphere :-



$S_1 \to x_1^2 + y_1^2 + z_1^2 + 2u_1x_1 + 2v_1y_1 + 2w_1z_1 + d_1 = 0$

$S_2 \to x_2^2 + y_2^2 + z_2^2 + 2u_2x_2 + 2v_2y_2 + 2w_2z_2 + d_2 = 0$

$$S_1 - S_2 = 0$$

→ If S_1 and S_2 of two common points intersect section it will be satisfied $S_1 - S_2 = 0$.

→ Sphere touch a given circle:-

→ The sphere touch a circle as intersection of the sphere S_1 and S_2 is equal to zero then it gives :

$$S_1 + \lambda S_2 = 0 \quad \text{or} \quad S + \lambda P = 0$$

(Q) find a eq^n of a sphere having the circle $x^2+y^2+z^2+10y-4z = 8$ and $x+y+z = 3$. is a great circle.

→ Define :- When the plane passing through the centre of the sphere, the section is called great circle. The centre and radius of a great circle as the same of sphere eq^n.

Sol^n:- The given eq^n is

$$x^2+y^2+z^2+10y-4z = 8 \text{ and}$$

$$x+y+z = 3$$

we know that

$$S + \lambda P = 0$$

$$(x^2+y^2+z^2+10y-4z-8)+\lambda(x+y+z-3)=0$$

$$x^2+y^2+z^2+10y-4z-8+x\lambda+y\lambda+z\lambda-3\lambda=0$$

$$x^2+y^2+z^2+(10+\lambda)y-(4-\lambda)z+x\lambda-3\lambda-8=0 \quad (1)$$

$\therefore$ center of circle is $=(-u,-v,-w)$

$$u=\frac{-\lambda}{2}, \quad v=\frac{10+\lambda}{2}, \quad w=\frac{4-\lambda}{2}$$

put value of u, v, w in

$$x+y+z=3$$

$$\frac{-\lambda}{2}+\frac{10+\lambda}{2}+\frac{4-\lambda}{2}=3$$

$$-\lambda-10-\lambda-\lambda+4=6$$

$$-3\lambda=6-4+10$$

$$-3\lambda=12$$

$$\boxed{\lambda=-4}$$

$$-\lambda+10+\lambda+4-\lambda=6$$

$$-3\lambda=6-14$$

$$-3\lambda=-8$$

$$\boxed{\cancel{\lambda=\frac{8}{3}}}$$

→ put value of λ in eq^n (1)

$$x^2+y^2+z^2-\left(10-\frac{8}{3}\right)y-\left(4-\frac{8}{3}\right)z+\frac{8x}{3}-\frac{24}{3}-8=0$$

$$x^2+y^2+z^2-\frac{22}{3}y-\frac{4}{3}z+\frac{8x}{3}-16=0$$

$$\boxed{x^2+y^2+z^2+6y-8z-4x-20=0}$$

(Q) Prove that the circle $x^2+y^2+z^2-2x+3y+4z-5$, $5y+6z+1=0$ and $x^2+y^2+z^2-3x-4y+5z-6=0$, $x+2y-7z=0$ lie on the sphere. Find its eqn.

Solh:-

$$S+\lambda P=0$$

$$x^2+y^2+z^2-2x+3y+4z-5+\lambda_1(5y+6z+1)=0$$

$$x^2+y^2+z^2-2x+3y+4z-5+5\lambda_1 y+6\lambda_1 z+\lambda_1=0 \quad -(1)$$

$$x^2+y^2+z^2-2x+(3+5\lambda_1)y+(4+6\lambda_1)z+\lambda_1-5=0 \quad (3)$$

~~u =~~

~~centre~~ of circle $=(-u,-v,-w)$

$$=(2,-(3+5\lambda_1),-(4+6\lambda_1))$$

$$u=\frac{2}{2},\quad v=\frac{(3+5\lambda_1)}{2},\quad w=\frac{(4+6\lambda_1)}{2}$$

$$S + \lambda P = 0$$

$$x^2+y^2+z^2-3x-4y+5z-6+\lambda_2(x+2y-7z)=0 \quad -②$$

$$x^2+y^2+z^2+(\lambda_2-3)x+(2\lambda_2-4)y+(5-7\lambda_2)z-6=0 \quad -0$$

→ These two sphere are identical

$$-2=\lambda_2-3, \quad 3+5\lambda_1=2\lambda_2-4, \quad 4+6\lambda_1=5-7\lambda_2$$

$$\lambda_2=1, \quad \lambda_1=-1$$

$$\boxed{x^2+y^2+z^2-2x-2y-2z-6=0}$$

$$\boxed{x^2+y^2+z^2-2x-2y-2z-6=0}$$

which is the required eqn.

Intersection of sphere and Line :-

→ If two value of r are real line cut the sphere in two points. If two value of r are co-incident line touch the sphere at one point. If two value of r are imaginary then line does not intersect the sphere.

$$x^2+y^2+z^2+2ux+2vy+2wz+d=0 \quad -(1)$$

centre

radius $= \sqrt{u^2+v^2+w^2-d}$

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1} = t$$

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} = r$$

$$\frac{x-x_1}{l} = r \qquad \frac{y-y_1}{m} = r \qquad \frac{z-z_1}{n} = r$$

$$\boxed{x = x_1 + lr} \qquad \boxed{y = y_1 + mr} \qquad \boxed{z = z_1 + nr} \quad -(2)$$

$(x_1 + lr_1)$ $\qquad$ $(x_1 + lr_2)$

$(y_1 + mr_1)$ $\qquad$ $(y_1 + mr_2)$

$(z_1 + nr_1)$ $\qquad$ $(z_1 + nr_2)$

⇒ Tangent line and Tangent plane of a sphere.

→ The line which intersect the sphere in two co-incident points is called Tangentline to the sphere.

→ The locus made by the tangent line at a point of the sphere is called tangent plane.

⇒ Eqⁿ of the tangent plane to the sphere:-

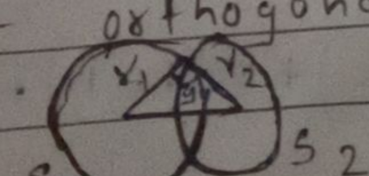
→ The eqⁿ of the tangent plane to the sphere $x^2+y^2+z^2+2ux+2vy+2wz+d=0$ at a point (x_1, y_1, z_1) is given as $x(x_1+u)+y(y_1+v)+z(z_1+w)+ux_1+vy_1+wz_1+d=0$

⇒ Condition of tangency:-

→ To find the condition of tangency that the plane $lx+my+nz=p$ then it can be touch with a eqⁿ of sphere. Such that perpendicular distance from the centre $(-u,-v,-w)$ to the tangent plane $lx+my+nz-p=0$ is equal to radius of the sphere. (r)

$$\boxed{\therefore \frac{-lx-my-nz-p}{\sqrt{l^2+m^2+n^2}} = \sqrt{u^2+v^2+w^2-d}}$$

⇒ Condition of orthogonality of two sphere:-



→ If two sphere S_1 and S_2

$S_1 = (-u_1, -v_1, -w_1)$

$S_2 = (-u_2, -v_2, -w_2)$

then, $r_1 = \sqrt{u_1^2+v_1^2+w_1^2-d}$

$r_2 = \sqrt{u_2^2+v_2^2+w_2^2-d}$

$\ldots v_2 + 2w_1w_2 = d_1+d_2$

Q. find the eq^n of sphere which touch the sphere $x^2+y^2+z^2-x+3y+2z-3=0$ at point $(1,1,-1)$ and pass through the origin

Sol^n:- The eq^n of tangent plane is

$x(x_1+u)+y(y_1+v)+z(z_1+w)+ux_1+vy_1+wz_1+d=0$

→ at point $(1,1,-1)$
x_1, y_1, z_1

$x(1+u)+y(1+v)+z(w-1)+u+v-w+d=0$ — (1)

given eq^n of sphere

$x^2+y^2+z^2-x+3y+2z-3=0$

$-u = 1/2$, $-v = -3/2$, $-w = -1$

centre $= (1/2, -3/2, -1)$

put value of u, v, w in eq^n - (1)

~~$x(1+\frac{1}{2})+y(1+3/2)+z(+1-1)+\frac{1}{2}+\frac{3}{2}-1+d=0$~~

~~$x(\frac{3}{2})+y(\frac{-1}{2})-2z-2+d=0$~~

~~$3x-y-z-1+2d=0$~~

$x(1-\frac{1}{2})+y(1+\frac{3}{2})+z(1-1)-\frac{1}{2}+\frac{3}{2}-1+d=0$

$x\cdot\frac{1}{2}+y\cdot\frac{5}{2}-\frac{1}{2}+\frac{3}{2}-1-3=0$

$x+5y-1+3-2-6=0$

$\boxed{x+5y-6=0}$

we know that.

$S + \lambda P = 0$

$(x^2+y^2+z^2-x+3y+2z-3) + \lambda(x+5y-6) = 0$

$x^2+y^2+z^2+(\lambda-1)x+(5\lambda+3)y+2z-6\lambda-3=0$

at origin $(0,0,0)$

$\therefore -3-6\lambda=0 \Rightarrow \lambda = -\frac{1}{2}$.

$(x^2+y^2+z^2-x+3y+2z-3) - \frac{1}{2}(x+5y-6) = 0$

$\cancel{2(x^2+y^2+z^2-x+3y}$

$\cancel{2x^2+2y^2}$

$2(x^2+y^2+z^2) - 2x + 6y + 4z - 6 - x - 5y + 6 = 0$

$2(x^2+y^2+z^2) - 3x + y + 4z = 0$
which is the req. eq^n of sphere.

(Q) find the eq^n of sphere which touch the sphere $x^2+y^2+z^2+2x-4y+6z-7=0$ at point $(1,1,-1)$ and pass through the origin.

→ The eq^n of tangent plane is
$x(x_1+u)+y(y_1+v)+z(z_1+w)+ux_1+y_1v+z_1w+d=0$

→ at point $(1,1,-1)$
$\quad\quad\quad\;\; x_1, y_1, z_1$

$x(1+u)+y(1+v)+z(w-1)+u+v-w+d=0$ — (1)

given eq^n of sphere
$x^2+y^2+z^2+2x-4y+6z-7=0$

$-u = -1$, $-v = 2$, $-w = -3$
$u = 1$ $v = -2$ $w = 3$
centre $= (-1, 2, -3)$

put value u, v, w in eq^n - 1

$x(1+1) + y(1-2) + z(3-1) + 1 - 2 - 3 + d = 0$

$2x - y + 2z - 4 - 7 = 0$
$\boxed{\therefore 2x - y + 2z - 11 = 0}$

we know that
$S + \lambda P = 0$
$x^2 + y^2 + z^2 + 2x - 4y + 6z - 7 + \lambda(2x - y + 2z - 11) = 0$

at origin $(0, 0, 0)$
$-7 - 11\lambda = 0$
$-11\lambda = 7$
$\boxed{\lambda = -\frac{7}{11}}$

$x^2 + y^2 + z^2 + 2x - 4y + 6z - 7 + \left(\frac{7}{11}\right)(2x - y + 2z - 11) = 0$

$x^2 + y^2 + z^2 + 2x - 4y + 6z - \cancel{7} - \frac{14x}{11} - \frac{7y}{11} - \frac{14z}{11} + \cancel{7} = 0$

$11x^2 + 11y^2 + 11z^2 + 22x - 44y + 66z - 14x - 7y - 14z = 0$

$\therefore 11(x^2 + y^2 + z^2) + 8x - 51y + 52z = 0$

which is required eq^n.

(9) find the eq^n of sphere that pass through the circle $x^2+y^2+z^2-2x+3y-4z+6=0$, $3x-4y+5z-15=0$ cut the sphere $x^2+y^2+z^2+2x+4y-6z+11=0$ orthogonality.

Sol^n:- We know that the given eq^n of of circle. pass the line is given as

$$S+\lambda P=0$$

$$x^2+y^2+z^2-2x+3y-4z+6+\lambda(3x-4y+5z-15)=0$$

$$x^2+y^2+z^2-(2-3\lambda)x+(3-4\lambda)y-(4-5\lambda)z+6-15\lambda=0 \quad (1)$$

But given eq^n of sphere cut the plane is

$$x^2+y^2+z^2+2x+4y-6z+11=0 \quad - (2)$$

Here,

$$u_1=-\frac{(2-3\lambda)}{2}, \quad v_1=\frac{3-4\lambda}{2}, \quad w_1=\frac{-(4-5\lambda)}{2}$$

$$u_2=1, \quad v_2=2, \quad w_2=-3$$

eq^n of ~~oth~~ orthogonality is given as

$$2u_1u_2+2v_1v_2+2w_1w_2=d_1+d_2$$

$$\cancel{2}\left(-\left(\frac{2-3\lambda}{\cancel{2}}\right)\right)\times 1+\cancel{2}\left(\frac{3-4\lambda}{\cancel{2}}\right)\cdot 2 \; \cancel{+}\; \cancel{2}\left(\frac{4-5\lambda}{\cancel{2}}\right)\cdot 3=6-15\lambda+11$$

$$-2+3\lambda+6-8\lambda+12-\cancel{15}\lambda=6-\cancel{15}\lambda+11$$

$$-5\lambda+16=17$$

$$-5\lambda=1$$

$$\boxed{\lambda=\frac{-1}{5}}$$

$$x^2+y^2+z^2-2x+3y-4z+6-\frac{1}{5}(3x+4y+5z-15)=0$$

$$5(x^2+y^2+z^2)-10x+15y-20z+30-3x+4y-5z+15$$

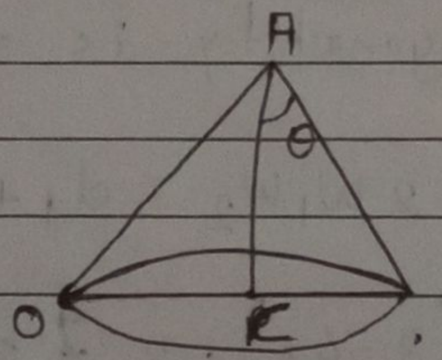
$$5(x^2+y^2+z^2)-13x+19y-25z+45=0$$

which is the required eq^n.

IMP ⇒ **Right Circular Cone :-**

→ A right circular cone is a surface generated by the straight line pass through if fixed point A and making a constant angle θ with a given line AC.

→ The fix point is called vertex. The constant angle θ is called semivertical angle and the fix line is called the axis of the cone.



→ Eq^n of the right circular cone :-

→ If the eq^n of right circular cone with vertics at the point A(α, β, γ), semi vertical angle θ and whose axes is the line AC is given as

$$\frac{x-\alpha}{l}=\frac{y-\beta}{m}=\frac{z-\gamma}{n}$$

case(i)

case(ii)

then eq^n of right circular cone is

$$\cos\theta = \frac{l(x-\alpha)+m(y-\beta)+n(z-\gamma)}{\sqrt{l^2+m^2+n^2}\cdot\sqrt{(x-\alpha)^2+(y-\beta)^2+(z-\gamma)^2}}$$

OR

$$(l^2+m^2+n^2)\{(x-\alpha)^2+(y-\beta)^2+(z-\gamma)^2\}\cos^2\theta = \left(l(x-\alpha)+m(y-\beta)+n(z-\gamma)\right)^2$$

case (i) If the vertics is at the origin $(0,0,0)$:-

$$\cos\theta = \frac{lx+my+nz}{\sqrt{l^2+m^2+n^2}\cdot\sqrt{x^2+y^2+z^2}}$$

$$\cos^2\theta\cdot(l^2+m^2+n^2)\cdot(x^2+y^2+z^2) = (lx+my+nz)^2$$

case (ii) If the vertices is at the origin and the z-axis is the axis of the cone then put $l=0, m=0, n=0$. such that eq^n is given as

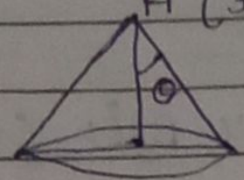
$$z^2\tan^2\theta = x^2+y^2$$

Date
Page

Q. find the eq^n of Right circular cone whose vertics 3,2,1, axis is the line $\frac{x-3}{4}=\frac{y-2}{1}=\frac{z-1}{3}$

and semivertical angle is $\frac{\pi}{6}$.

Ans:-



we given vertics of a right circular cone (3,2,1), $\theta=\frac{\pi}{6}$ and axis line $\frac{x-3}{4}=\frac{y-2}{1}=\frac{z-1}{3}$.

we comparing with

$$\frac{x-\alpha}{l}=\frac{y-\beta}{m}=\frac{z-\gamma}{n}$$

$\therefore l = 4$	$\alpha = 3$
$m = 1$	$\beta = 2$
$n = 3$	$\gamma = 1$

eq^n of right circular cone.

$$\cos\theta = \frac{l(x-\alpha)+m(y-\beta)+n(z-\gamma)}{\sqrt{l^2+m^2+n^2}\sqrt{(x-\alpha)^2+(y-\beta)^2+(z-\gamma)^2}}$$

$$\cos\frac{\pi}{6} = \frac{4(x-3)+1(y-2)+3(z-1)}{\sqrt{16+1+9}\cdot\sqrt{(x-3)^2+(y-2)^2+(z-1)^2}}$$

$$\frac{\sqrt{3}}{2} = \frac{4(x-3)+1(y-2)+3(z-1)}{\sqrt{26}\sqrt{(x-3)^2+(y-2)^2+(z-1)^2}}$$

$\rightarrow \dfrac{9}{4} = \dfrac{[4(x-3)+(y-2)+3(z-1)]^2}{26\,[(x-3)^2+(y-2)^2+(z-1)^2]}$

$\rightarrow \dfrac{9}{4}\cdot 26\,[(x-3)^2+(y-2)^2+(z-1)^2] = [4(x-3)+(y-2)+3(z-1)]^2$

$\rightarrow \dfrac{39}{2}[x^2-6x+9+y^2-4y+4+z^2-2z+1] = [(4x-12)+(y-2)+(3z-3)]^2$

$\rightarrow \dfrac{39}{2}[x^2+y^2+z^2-6x-4y-2z+14] = (4x-12)^2$

$16x^2-96x+144+y^2-4y+4+9z^2-18z+9+$

$\cancel{2(y-4x}$

$2\underbrace{(4x-12)(y-2)}_{8x-24}+2\underbrace{(y-2)(3z-3)}_{2y-4}+2\underbrace{(3z-3)(4x-12)}_{6z-6}$

$\rightarrow \dfrac{39}{2}[x^2+y^2+z^2-6x-4y-2z+14] = 16x^2+y^2+9z^2-96x-4y-$

$18z+157+8xy-16x-24y+48+6yz-6y$

$-12z+12+24xz-72z-24x+72$

$\rightarrow \dfrac{39}{2}[x^2+y^2+z^2-6x-4y-2z+14] = 16x^2+y^2+9z^2-136x-34y-102z$

$+289+8xy+6yz+24xz$

$\rightarrow 39x^2+39y^2+39z^2-234x-156y-78z+546 = 32x^2+$

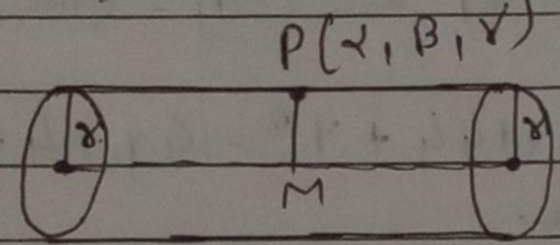
$2y^2+18z^2-272x-68y-204z+578+$

$16xy+12yz+48xz$

$\rightarrow$ $\boxed{7x^2+37y^2+21z^2+38x-88y+126z-32+16xy+12yz+48xz}$ which is req eqn.

Right circular cylinder :-

→ A right circular cylinder is a surface generated by a straight line which remains parallel to a fixed line and is at a constant distance from it. The fixed line is called axis and the constant distance is called radius, (r) of the cylinder. The plane section of a right circular cylinder perpendicular to its axis is called normal section.



→ Eqn of right circular cylinder.

→ If radius of circular cylinder (r) and axis line is

$$\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$$

→ Let P is the point of the surface.
Let P(α, β, γ) be any point on the surface of cylinder we draw a perpendicular line PM on the axis then eqn of right circular cylinder is

$$(x-\alpha)^2+(y-\beta)^2+(z-\gamma)^2 = r^2 + \frac{[l(x-\alpha)+m(y-\beta)+n(z-\gamma)]^2}{l^2+m^2+n^2}$$

case

case (i) If axis of the cylinder is x axis then
$l=1,\ m=0,\ n=0$

$$(x-\alpha)^2+(y-\beta)^2+(z-\gamma)^2=r^2+\left[\frac{l(x-\alpha)}{l^2}\right]^2$$

Q Find the eq^n of right circular cylinder of radius 2, whose axis is the line
$$\frac{(x-1)}{2}=\frac{y-2}{1}=\frac{z-3}{2} \quad -①$$

→ eq^n. of R.C.C.

$$(x-\alpha)^2+(y-\beta)^2+(z-\gamma)^2=r^2+\frac{[l(x-\alpha)+m(y-\beta)+n(z-\gamma)]^2}{l^2+m^2+n^2} \quad -②$$

comparing with $\frac{x-\alpha}{l}=\frac{y-\beta}{m}=\frac{z-\gamma}{n}$ of eq^n-①

$\alpha=1,\ \beta=2,\ \gamma=3$
$l=2,\ m=1,\ n=2$

put in eq^n - ②

$$(x-1)^2+(y-2)^2+(z-3)^2=(2)^2+\frac{[2(x-1)+1(y-2)+2(z-3)]^2}{2^2+1^2+2^2}$$

$$(x-1)^2+(y-2)^2+(z-3)^2=4+\frac{[(2x-2)+(y-2)+(2z-6)]^2}{4+1+4}$$

$$9[(x-1)^2+(y-2)^2+(z-3)^2]=36+4x^2-8x+4+y^2-4y+4+$$
$$4z^2-24z+36+2[2xy-4x-2y+4]+2[2zy-6y-4z+12]$$
$$+2[4xz-12x-4z+12]$$

$\rightarrow 9[x^2-2x+1+y^2-4y+4+z^2-6z+9] = 36+4x^2-8x+4+$

$y^2-4y+4+4z^2-24z+36+4xy-8x-4y+8$

$+4zy-12y-8z+24+8xz-24x-8z+24$

$\rightarrow 9x^2-18x+9+9y^2-36y+36+9z^2-54z+81 = 4x^2+y^2+$

$+4z^2-40x-20y-40z+4xy+4zy+8xz+136$

$\rightarrow 5x^2+8y^2+5z^2+22x-16y-14z-10-4xy-4zy$
$-8xz.$

$\therefore 5x^2+8y^2+5z^2-4xy-4zy-8xz+22x-16y-14z-10$
which is required eq^n.